

Introduction to Data Science and Engineering

- Importing data



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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Some slides retrofitted by Michael Wu



Why importing data?

- So far, we have worked on built-in datasets. But real-life datasets are everything, everywhere, and come in all kinds of file formats
- Common data file formats
 - `csv`, Comma-Separated Values;
 - `tsv`, Tab-Separated Values;
 - `xls/xlsx`, Microsoft Excel worksheets;
 - `txt`, `json`, `html` : other text-based files;
 - `parquet`, `feather`, `hdf5`, `zarr`: binary data files.
- These formats can be imported to R as data frame easily

An example csv file

```
1 ACQUISITION_ID,SAMPLE_LABEL,SAMPLE_ID,tissue_type,tissue_subtype,diagnosis,normal or neoplasia,Full_Histology,Full_Tumor_Location,Simple_Tumor_Location,PATIENT_ID,pN,pM,V,OS,species,treatment,Sex,Age,LA_1,Diffuse_1,LA_2,
2 Coordina-14_c001_v001_r001_reg002,TMA-A_reg002,24527.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68285.0,1c,,0,71.335,human,,Female,76.1,1,1,1,1,2,2,0,0,3,3B,0,3,2,1,0,71.335,0,25,3,1,1,1,1,MSS,0
3 Coordina-14_c001_v001_r001_reg004,TMA-A_reg004,24529.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68300.0,1b,,0,137.64,human,,Female,85.29,,2,2,2,0,0,0,3,3B,0,3,2,1,0,137.64,1,8,2,1,1,1,1,MSS,0
4 Coordina-14_c001_v001_r001_reg005,TMA-A_reg005,24530.0,Descendens,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1,0.394,human,,Male,56.01,,2,2,2,0,0,0,3,3B,0,3,2,1,0.394,0,5,2,1,1,1,1,MSS,0
5 Coordina-14_c001_v001_r001_reg006,TMA-A_reg006,24531.0,Descendens,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1,0.394,human,,Male,56.01,,2,2,2,0,0,0,3,3B,0,3,2,1,0.394,0,5,2,1,1,1,1,MSS,0
6 Coordina-14_c001_v001_r001_reg007,TMA-A_reg007,24532.0,Sigma,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1,0.526,human,,Male,82.36,,2,2,2,0,0,0,3,3B,0,3,2,1,0.526,0,0,1,0,1,1,MSI,0
7 Coordina-14_c001_v001_r001_reg008,TMA-A_reg008,24533.0,Sigma,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1,0.526,human,,Male,82.36,,2,2,2,0,0,0,3,3B,0,3,2,1,0.526,0,0,1,0,1,1,MSI,0
8 Coordina-14_c001_v001_r001_reg009,TMA-A_reg009,24534.0,Transversum,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0,43.458,human,,Female,78.38,,2,2,2,0,0,0,3,3B,0,3,2,1,0.43.458,0,12,3,1,1,1,1,MSS,0
9 Coordina-14_c001_v001_r001_reg010,TMA-A_reg010,24535.0,Transversum,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0,43.458,human,,Female,78.38,,2,2,2,0,0,0,3,3B,0,3,2,1,0.43.458,0,12,3,1,1,1,1,MSS,0
10 Coordina-14_c001_v001_r001_reg011,TMA-A_reg011,24536.0,Ascendens,,,Mucinous,Ascendens,Right,68344.0,1b,,0,125.542,human,,Male,51.17,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
11 Coordina-14_c001_v001_r001_reg012,TMA-A_reg012,24537.0,Ascendens,,,Mucinous,Ascendens,Right,68344.0,1b,,0,125.542,human,,Male,51.17,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
12 Coordina-14_c001_v001_r001_reg013,TMA-A_reg013,24538.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0,68.179,human,,Male,70.58,,2,2,2,0,0,0,3,3B,0,3,2,1,0,68.179,1,6,2,1,1,1,1,MSS,0
13 Coordina-14_c001_v001_r001_reg014,TMA-A_reg014,24539.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0,68.179,human,,Male,70.58,,2,2,2,0,0,0,3,3B,0,3,2,1,0,68.179,1,6,2,1,1,1,1,MSS,0
14 Coordina-14_c001_v001_r001_reg015,TMA-A_reg015,24540.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0,21.236,human,,Male,30.37,,2,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
15 Coordina-14_c001_v001_r001_reg016,TMA-A_reg016,24541.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0,21.236,human,,Male,30.37,,2,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
16 Coordina-14_c001_v001_r001_reg017,TMA-A_reg017,24542.0,Cecum,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1,12.788,human,,Female,74.29,,2,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
17 Coordina-14_c001_v001_r001_reg018,TMA-A_reg018,24543.0,Cecum,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1,12.788,human,,Female,74.29,,2,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
18 Coordina-14_c001_v001_r001_reg019,TMA-A_reg019,24544.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,1,124.096,human,,Male,61.68,1,1,1,1,2,2,0,1,2,3A,0,3,2,0,124.096,0,6,2,1,1,1,1,MSS,0
19 Coordina-14_c001_v001_r001_reg020,TMA-A_reg020,24545.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,1,124.096,human,,Male,61.68,1,1,1,1,2,2,0,1,2,3A,0,3,2,0,124.096,0,6,2,1,1,1,1,MSS,0
20 Coordina-14_c001_v001_r001_reg021,TMA-A_reg021,24546.0,Descendens,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1,70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
21 Coordina-14_c001_v001_r001_reg022,TMA-A_reg022,24547.0,Descendens,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1,70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
22 Coordina-14_c001_v001_r001_reg023,TMA-A_reg023,24548.0,Sigma,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0,105.161,human,,Female,75.87,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
23 Coordina-14_c001_v001_r001_reg024,TMA-A_reg024,24549.0,Sigma,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0,105.161,human,,Female,75.87,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
24 Coordina-14_c001_v001_r001_reg025,TMA-A_reg025,24550.0,Cecum,,,Mucinous,Cecum,Right,68293.0,1b,,0,42.965,human,,Female,55.35,1,1,1,1,2,2,0,1,3,3B,0,3,3,1,0,42.965,0,3,1,1,nd,nd,MSS,0
25 Coordina-14_c001_v001_r001_reg027,TMA-A_reg027,24552.0,Cecum,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,2,3,0,0,0,3,4,1,4,3,1,0,13.248,0,13,3,1,1,1,1,MSS,0
26 Coordina-14_c001_v001_r001_reg028,TMA-A_reg028,24553.0,Cecum,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,2,3,0,0,0,3,4,1,4,3,1,0,13.248,0,13,3,1,1,1,1,MSS,0
27 Coordina-14_c001_v001_r001_reg029,TMA-A_reg029,24554.0,Sigma,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,1,0.296,human,,Female,66.01,,2,2,2,0,0,0,3,4,1,4,2,1,0.296,0,4,1,1,nd,nd,MSS,0
28 Coordina-14_c001_v001_r001_reg030,TMA-A_reg030,24555.0,Sigma,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,1,0.296,human,,Female,66.01,,2,2,2,0,0,0,3,4,1,4,2,1,0.296,0,4,1,1,nd,nd,MSS,0
29 Coordina-14_c001_v001_r001_reg031,TMA-A_reg031,24556.0,Cecum,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1,124.622,human,,Female,77.24,,2,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
30 Coordina-14_c001_v001_r001_reg032,TMA-A_reg032,24557.0,Cecum,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1,124.622,human,,Female,77.24,,2,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
31 Coordina-14_c001_v001_r001_reg033,TMA-A_reg033,24558.0,Cecum,,,Mucinous,Cecum,Right,68297.0,1b,,0,138.034,human,,Male,63.16,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,138.034,0,9,2,1,1,1,1,MSS,0
32 Coordina-14_c001_v001_r001_reg036,TMA-A_reg036,24561.0,Sigma,,,Adenocarcinoma,Sigma,Left,68298.0,2b,1,1,11.933,human,,Male,60.78,1,1,2,2,0,0,1,4,0,4,3,1,1,10.552,1,10,3,nd,nd,nd,nd,
33 Coordina-14_c001_v001_r001_reg037,TMA-A_reg037,24562.0,Cecum,,,Mucinous,Cecum,Right,68299.0,1c,,0,17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,MSS,0
34 Coordina-14_c001_v001_r001_reg038,TMA-A_reg038,24563.0,Cecum,,,Mucinous,Cecum,Right,68299.0,1c,,0,17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,MSS,0
35 Coordina-14_c001_v001_r001_reg044,TMA-A_reg044,24569.0,Descendens,,,Adenocarcinoma,Descendens,Left,68303.0,2a,,0,37.9,human,,Female,89.53,2,,2,2,0,0,4,3C,3,2,0,1,0,1,6,2,1,nd,nd,MSS,0
36 Coordina-14_c001_v001_r001_reg045,TMA-A_reg045,24570.0,Cecum,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0,35.574,human,,Male,29.87,,2,2,2,0,0,0,3,3B,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,0
37 Coordina-14_c001_v001_r001_reg046,TMA-A_reg046,24571.0,Cecum,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0,35.574,human,,Male,29.87,,2,2,2,0,0,0,3,3B,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,0
38 Coordina-14_c001_v001_r001_reg047,TMA-A_reg047,24572.0,Cecum,,,Mucinous,Cecum,Right,68305.0,1a,,1,32.721,human,,Female,50.98,1,1,1,1,2,0,0,3,3B,3,3,1,0,32.721,0,2,1,1,1,1,MSS,0
39 Coordina-14_c001_v001_r001_reg048,TMA-A_reg048,24573.0,Cecum,,,Mucinous,Cecum,Right,68305.0,1a,,1,32.721,human,,Female,50.98,1,1,1,1,2,0,0,3,3B,3,3,1,0,32.721,0,2,1,1,1,1,MSS,0
40 Coordina-14_c001_v001_r001_reg052,24577.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68307.0,2b,,1,0.559,human,,Female,80.52,2,,2,2,0,0,3,3C,0,3,2,1,0.559,0,3,1,1,1,1,MSS,0
41 Coordina-14_c001_v001_r001_reg053,TMA-A_reg053,24578.0,Sigma,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0,1.282,human,,Male,81.2,,2,2,2,0,0,0,4a,3B,0,3,2,1,1.282,0,5,2,1,1,1,1,MSS,0
42 Coordina-14_c001_v001_r001_reg054,TMA-A_reg054,24579.0,Sigma,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0,1.282,human,,Male,81.2,,2,2,2,0,0,0,4a,3B,0,3,2,1,1.282,0,5,2,1,1,1,1,MSS,0
43 Coordina-14_c001_v001_r001_reg061,24586.0,Ascendens,,,Mucinous,Ascendens,Right,68313.0,1a,,1,10.355,human,,Male,86.44,,2,2,2,0,0,0,3,3B,0,3,1,1,10.355,0,3,1,0,0,1,MSI,0
44 Coordina-14_c001_v001_r001_reg062,TMA-A_reg062,24587.0,Ascendens,,,Mucinous,Ascendens,Right,68313.0,1a,,1,10.355,human,,Male,86.44,,2,2,2,0,0,0,3,3B,0,3,1,1,10.355,0,3,1,0,0,1,MSI,0
45 Coordina-14_c001_v001_r001_reg064,TMA-A_reg064,24589.0,Sigma,,,Adenocarcinoma,Sigma,Left,68314.0,2b,1,1,11.275,human,,Female,59.67,1,1,1,1,2,2,0,0,4,4,0,4,2,1,0,1,6,2,1,1,1,1,MSS,0
46 Coordina-14_c001_v001_r001_reg065,TMA-A_reg065,24590.0,Sigma,,,Adenocarcinoma,Sigma,Left,68315.0,0,0,1,1,16.2,human,,Female,79.49,1,1,1,1,2,0,0,4a,4,4,0,1,16.2,0,3,1,1,1,1,MSS,0
47 Coordina-14_c001_v001_r001_reg066,TMA-A_reg066,24591.0,Sigma,,,Adenocarcinoma,Sigma,Left,68315.0,0,0,1,1,16.2,human,,Female,79.49,1,1,1,1,2,0,0,4a,4,4,0,1,16.2,0,3,1,1,1,1,MSS,0
48 Coordina-14_c001_v001_r001_reg067,TMA-A_reg067,24592.0,Descendens,,,Adenocarcinoma,Descendens,Left,68316.0,0,0,1,0,18.54,human,,Male,68.05,1,1,1,1,2,2,0,0,4a,4,4,2,1,0,18.54,1,5,2,1,1,1,1,MSS,
```

A dataset of colorectal cancer patients, each row represents a patient, with:

- Tumor type, location, grading and stage, demographics, etc.

An example csv file

```
1 ACQUISITION_ID,SAMPLE_LABEL,SAMPLE_ID,tissue_type,tissue_subtype,diagnosis,normal_or_neoplasia,Full_Histology,Full_Tumor_Location,Simple_Tumor_Location,PATIENT_ID,pN,pM,V,OS,species,treatment,Sex,Age,LA_1,Diffuse_1,LA_2,
2 Coordina-14_c001_v001_r001_reg002,TMA-A_reg002,24527.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68333.0,1a,,0.71.335,human,,Female,76.1,1,1,1,1,2,2,0,0,3,3B,0,3,2,1,0,71.335,0,25,3,1,1,1,1,MSS,0
3 Coordina-14_c001_v001_r001_reg004,TMA-A_reg004,24529.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68300.0,1b,,0.137.64,human,,Female,85.29,,2,2,2,0,0,0,3,3B,0,3,2,1,0,137.64,1,8,2,1,1,1,1,MSS,0
4 Coordina-14_c001_v001_r001_reg005,TMA-A_reg005,24530.0,Descendens,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1.0.394,human,,Male,56.01,,2,2,2,0,0,0,3,3B,0,3,2,1,0.394,0,5,2,1,1,1,1,MSS,0
5 Coordina-14_c001_v001_r001_reg006,TMA-A_reg006,24531.0,Descendens,,,Adenocarcinoma,Descendens,Left,68311.0,1b,,1.0.394,human,,Male,56.01,,2,2,2,0,0,0,3,3B,0,3,2,1,0.394,0,5,2,1,1,1,1,MSS,0
6 Coordina-14_c001_v001_r001_reg007,TMA-A_reg007,24532.0,Sigma,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1.0.526,human,,Male,82.36,,2,2,2,0,0,0,3,3B,0,3,2,1,0.526,0,0,1,1,0,1,1,MSI,0
7 Coordina-14_c001_v001_r001_reg008,TMA-A_reg008,24533.0,Sigma,,,Adenocarcinoma,Sigma,Left,68322.0,1a,,1.0.526,human,,Male,82.36,,2,2,2,0,0,0,3,3B,0,3,2,1,0.526,0,0,1,1,0,1,1,MSI,0
8 Coordina-14_c001_v001_r001_reg009,TMA-A_reg009,24534.0,Transversum,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0.43.458,human,,Female,78.38,,2,2,2,0,0,0,3,3B,0,3,2,1,0.43.458,0,12,3,1,1,1,1,MSS,0
9 Coordina-14_c001_v001_r001_reg010,TMA-A_reg010,24535.0,Transversum,,,Adenocarcinoma,Transversum,Right,68333.0,1a,,0.43.458,human,,Female,78.38,,2,2,2,0,0,0,3,3B,0,3,2,1,0.43.458,0,12,3,1,1,1,1,MSS,0
10 Coordina-14_c001_v001_r001_reg011,TMA-A_reg011,24536.0,Ascendens,,,Mucinous,Ascendens,Right,68344.0,1b,,0.125.542,human,,Male,51.17,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
11 Coordina-14_c001_v001_r001_reg012,TMA-A_reg012,24537.0,Ascendens,,,Mucinous,Ascendens,Right,68344.0,1b,,0.125.542,human,,Male,51.17,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,125.542,0,7,2,1,1,1,1,MSS,0
12 Coordina-14_c001_v001_r001_reg013,TMA-A_reg013,24538.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0.68.179,human,,Male,70.58,,2,2,2,0,0,0,3,3B,0,3,2,1,0.68.179,1,6,2,1,1,1,1,MSS,0
13 Coordina-14_c001_v001_r001_reg014,TMA-A_reg014,24539.0,Ascendens,,,Adenocarcinoma,Ascendens,Right,68355.0,1c,,0.68.179,human,,Male,70.58,,2,2,2,0,0,0,3,3B,0,3,2,1,0.68.179,1,6,2,1,1,1,1,MSS,0
14 Coordina-14_c001_v001_r001_reg015,TMA-A_reg015,24540.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0.21.236,human,,Male,30.37,,2,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
15 Coordina-14_c001_v001_r001_reg016,TMA-A_reg016,24541.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68366.0,2a,,0.21.236,human,,Male,30.37,,2,2,2,0,0,1,3,3B,1,4,2,1,1,21.236,1,21,3,1,1,1,1,MSS,0
16 Coordina-14_c001_v001_r001_reg017,TMA-A_reg017,24542.0,Cecum,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1.12.788,human,,Female,74.29,,2,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
17 Coordina-14_c001_v001_r001_reg018,TMA-A_reg018,24543.0,Cecum,,,Adenocarcinoma,Cecum,Right,68377.0,1b,,1.12.788,human,,Female,74.29,,2,2,2,0,0,0,4a,3B,0,3,3,1,1,12.788,1,5,2,0,0,1,1,MSI,1
18 Coordina-14_c001_v001_r001_reg019,TMA-A_reg019,24544.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,1.124.096,human,,Male,61.68,1,1,1,1,1,2,2,0,1,2,3A,0,3,2,1,0,124.096,0,6,2,1,1,1,1,MSS,0
19 Coordina-14_c001_v001_r001_reg020,TMA-A_reg020,24545.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68286.0,1b,,1.124.096,human,,Male,61.68,1,1,1,1,1,2,2,0,1,2,3A,0,3,2,1,0,124.096,0,6,2,1,1,1,1,MSS,0
20 Coordina-14_c001_v001_r001_reg021,TMA-A_reg021,24546.0,Descendens,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1.70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
21 Coordina-14_c001_v001_r001_reg022,TMA-A_reg022,24547.0,Descendens,,,Adenocarcinoma,Descendens,Left,68291.0,2a,,1.70.118,human,,Female,87.8,2,,1,1,1,2,0,0,3,3B,0,3,2,1,0,70.118,0,8,2,1,1,1,1,MSS,0
22 Coordina-14_c001_v001_r001_reg023,TMA-A_reg023,24548.0,Sigma,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0.105.161,human,,Female,75.87,1,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
23 Coordina-14_c001_v001_r001_reg024,TMA-A_reg024,24549.0,Sigma,,,Adenocarcinoma,Sigma,Left,68292.0,2b,,0.105.161,human,,Female,75.87,1,1,1,1,1,2,2,0,0,3,3C,0,3,2,1,0,105.161,0,16,3,1,1,1,1,MSS,0
24 Coordina-14_c001_v001_r001_reg025,TMA-A_reg025,24550.0,Cecum,,,Mucinous,Cecum,Right,68293.0,1b,,0.42.965,human,,Female,55.25,,2,2,2,0,0,1,3,3B,0,3,3,1,0,42.965,0,3,1,1,nd,nd,MSS,0
25 Coordina-14_c001_v001_r001_reg027,TMA-A_reg027,24552.0,Cecum,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,2,2,0,0,0,3,3B,0,3,3,1,1,13.248,0,13,3,1,1,1,1,MSS,0
26 Coordina-14_c001_v001_r001_reg028,TMA-A_reg028,24553.0,Cecum,,,Adenocarcinoma,Cecum,Right,68294.0,2b,1,1,13.248,human,,Male,56.63,,2,2,2,0,0,0,3,3B,0,3,3,1,1,13.248,0,13,3,1,1,1,1,MSS,0
27 Coordina-14_c001_v001_r001_reg029,TMA-A_reg029,24554.0,Sigma,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,1,0.296,human,,Female,66.01,,2,2,2,0,0,0,3,4,1,4,2,1,0.296,0,4,1,1,nd,nd,MSS,0
28 Coordina-14_c001_v001_r001_reg030,TMA-A_reg030,24555.0,Sigma,,,Adenocarcinoma,Sigma,Left,68295.0,2b,1,1,0.296,human,,Female,66.01,,2,2,2,0,0,0,3,4,1,4,2,1,0.296,0,4,1,1,nd,nd,MSS,0
29 Coordina-14_c001_v001_r001_reg031,TMA-A_reg031,24556.0,Cecum,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1.124.622,human,,Female,77.24,,2,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
30 Coordina-14_c001_v001_r001_reg032,TMA-A_reg032,24557.0,Cecum,,,Adenocarcinoma,Cecum,Right,68296.0,1b,,1.124.622,human,,Female,77.24,,2,2,2,0,0,1,2,3A,0,3,2,1,0,124.622,0,11,3,1,1,1,1,MSS,0
31 Coordina-14_c001_v001_r001_reg033,TMA-A_reg033,24558.0,Cecum,,,Mucinous,Cecum,Right,68297.0,1b,,0.138.034,human,,Male,63.16,1,1,1,1,1,1,2,0,1,3,3B,0,3,3,1,0,138.034,0,9,2,1,1,1,1,MSS,0
32 Coordina-14_c001_v001_r001_reg036,TMA-A_reg036,24561.0,Sigma,,,Adenocarcinoma,Sigma,Left,68298.0,2b,1,1,11.933,human,,Male,60.78,1,1,1,1,1,2,2,0,0,1,4,4,0,4,3,1,10.552,1,10,3,nd,nd,nd,nd,
33 Coordina-14_c001_v001_r001_reg037,TMA-A_reg037,24562.0,Cecum,,,Mucinous,Cecum,Right,68299.0,1c,,0.17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,MSS,0
34 Coordina-14_c001_v001_r001_reg038,TMA-A_reg038,24563.0,Cecum,,,Mucinous,Cecum,Right,68299.0,1c,,0.17.719,human,,Male,76.88,2,,1,1,1,2,0,0,3,3B,0,3,3,0,1,17.719,0,0,1,1,nd,nd,MSS,0
35 Coordina-14_c001_v001_r001_reg044,TMA-A_reg044,24569.0,Descendens,,,Adenocarcinoma,Descendens,Left,68303.0,2a,,0.37.9,human,,Female,89.53,2,,2,2,0,0,3,4,3C,0,3,2,0,1,0,6,2,1,nd,nd,MSS,0
36 Coordina-14_c001_v001_r001_reg045,TMA-A_reg045,24570.0,Cecum,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0.35.574,human,,Male,29.87,,2,2,2,0,0,0,3,3B,0,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,1
37 Coordina-14_c001_v001_r001_reg046,TMA-A_reg046,24571.0,Cecum,,,Adenocarcinoma,Cecum,Right,68304.0,1b,,0.35.574,human,,Male,29.87,,2,2,2,0,0,0,3,3B,0,3,2,1,0,35.574,0,12,3,1,1,0,0,MSI,1
38 Coordina-14_c001_v001_r001_reg047,TMA-A_reg047,24572.0,Cecum,,,Mucinous,Cecum,Right,68305.0,1a,,1.38.721,human,,Female,50.98,1,1,1,1,1,2,0,0,3,3B,0,3,3,1,0,38.721,0,2,1,1,1,1,MSS,0
39 Coordina-14_c001_v001_r001_reg048,TMA-A_reg048,24573.0,Cecum,,,Mucinous,Cecum,Right,68305.0,1a,,1.38.721,human,,Female,50.98,1,1,1,1,1,2,0,0,3,3B,0,3,3,1,0,38.721,0,2,1,1,1,1,MSS,0
40 Coordina-14_c001_v001_r001_reg052,TMA-A_reg052,24577.0,Rectum,,,Adenocarcinoma,Rectum,Rectum,68307.0,2b,,1.0.559,human,,Female,80.52,2,,2,2,0,0,3,3C,0,3,2,1,0,0.559,0,3,1,1,1,1,MSS,0
41 Coordina-14_c001_v001_r001_reg053,TMA-A_reg053,24578.0,Sigma,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0.1.282,human,,Male,81.2,2,,2,2,0,0,0,4a,3B,0,3,2,1,1,1.282,0,5,2,1,1,1,1,MSS,0
42 Coordina-14_c001_v001_r001_reg054,TMA-A_reg054,24579.0,Sigma,,,Adenocarcinoma,Sigma,Left,68308.0,1b,,0.1.282,human,,Male,81.2,2,,2,2,0,0,0,4a,3B,0,3,2,1,1,1.282,0,5,2,1,1,1,1,MSS,0
43 Coordina-14_c001_v001_r001_reg062,TMA-A_reg062,24586.0,Ascendens,,,Mucinous,Ascendens,Right,68313.0,1a,,1.10.355,human,,Male,86.44,2,,2,2,0,0,0,3,3B,0,3,1,1,1,10.355,0,3,1,0,1,1,MSI,1
44 Coordina-14_c001_v001_r001_reg062,TMA-A_reg062,24587.0,Ascendens,,,Mucinous,Ascendens,Right,68313.0,1a,,1.10.355,human,,Male,86.44,2,,2,2,0,0,0,3,3B,0,3,1,1,1,10.355,0,3,1,0,1,1,MSI,1
45 Coordina-14_c001_v001_r001_reg064,TMA-A_reg064,24589.0,Sigma,,,Adenocarcinoma,Sigma,Left,68315.0,0,1,1,16.2,human,,Female,59.67,1,1,1,1,1,2,2,0,0,4,4,0,4,2,1,0,1,6,2,1,1,1,1,MSS,0
46 Coordina-14_c001_v001_r001_reg065,TMA-A_reg065,24590.0,Sigma,,,Adenocarcinoma,Sigma,Left,68315.0,0,1,1,16.2,human,,Female,59.67,1,1,1,1,1,2,2,0,0,4,4,0,4,2,1,0,1,6,2,1,1,1,1,MSS,0
47 Coordina-14_c001_v001_r001_reg066,TMA-A_reg066,24591.0,Sigma,,,Adenocarcinoma,Sigma,Left,68315.0,0,1,1,16.2,human,,Female,59.67,1,1,1,1,1,2,2,0,0,4,4,0,4,2,1,0,1,6,2,1,1,1,1,MSS,0
48 Coordina-14_c001_v001_r001_reg067,TMA-A_reg067,24592.0,Descendens,,,Adenocarcinoma,Descendens,Left,68316.0,0,1,1,18.54,human,,Male,68.05,1,1,1,1,1,2,2,0,0,4,4,4,4,2,1,0,18.54,1,5,2,1,1,1,1,MSS,
```

Missing data

Text strings

This is tumor stage, should it be string? Integer? or factor?

Numeric (Age)

These are sample IDs, should they be numeric? or integer?

A dataset of colorectal cancer patients, each row represents a patient, with:

- Tumor type, location, grading and stage, demographics, etc.

murders dataset in a CSV file

The table loaded by `data(murders)` can be downloaded from <https://raw.githubusercontent.com/rafalab/dslabs/master/inst/extdata/murders.csv>

```
> library(dslabs)
> data(murders)
```

1	state,abb,region,population,total
2	Alabama,AL,South,4779736,135
3	Alaska,AK,West,710231,19
4	Arizona,AZ,West,6392017,232
5	Arkansas,AR,South,2915918,93
6	California,CA,West,37253956,1257
7	Colorado,CO,West,5029196,65
8	Connecticut,CT,Northeast,3574097,97
9	Delaware,DE,South,897934,38
10	District of Columbia,DC,South,601723,99

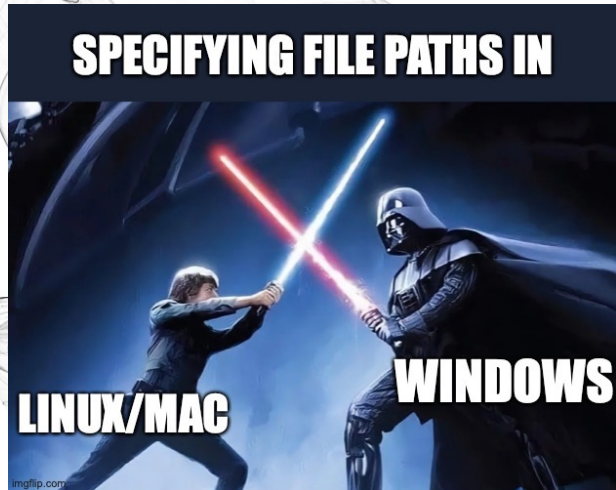
The first row (i.e., header) contains column names. Most data import functions assume presence of a header. How to check?

- In R: `read_lines("murders.csv", n_max = 3)`
- In shell: `head murders.csv -n 3`

Following rows contain actual table contents, note that `,` serve as delimiter between columns.

- Delimiters can also be other symbols such as semicolon `;`
- You can specify which delimiter to look for when parsing files.

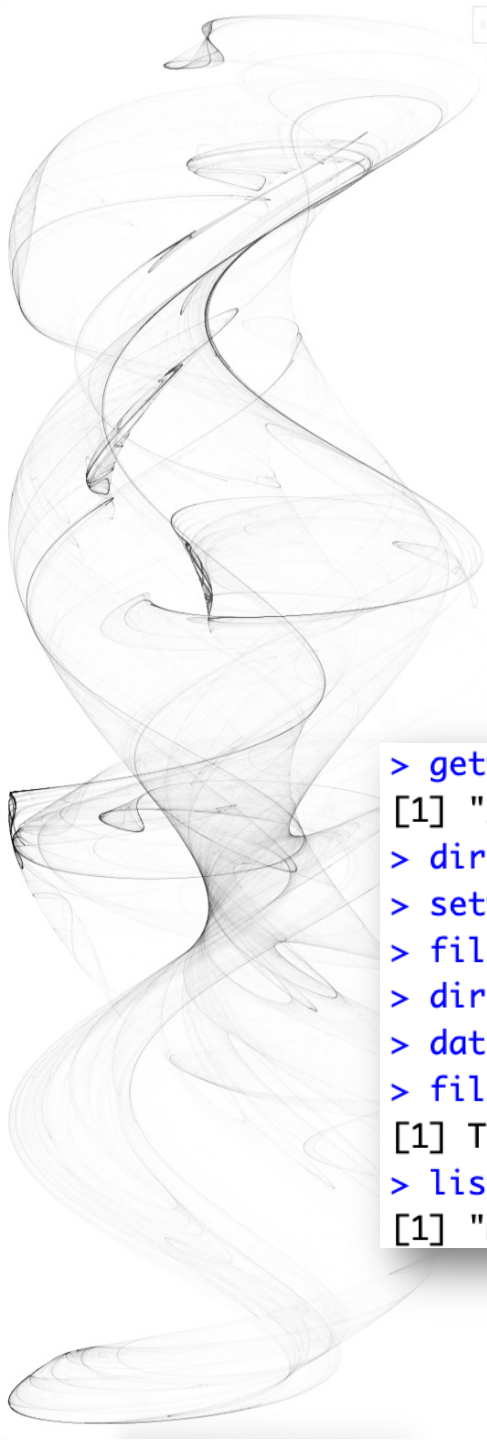
Relative and Absolute Paths



- Absolute path: the complete location of a file/folder from the root.
 - Linux/Mac: starting from '/'
 - Windows: starting from 'C: \', 'D: \', etc., note that forward slash (/) usually works for Windows paths (in Rstudio) as well;

```
> system.file(package = "dslabs")  
[1] "/Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/dslabs"
```

- Relative path: anything that doesn't start from the root
 - file_in_this_folder.txt
 - subdir/file_in_subdir.txt
 - ../file_in_parent_folder.txt
- In R:
 - file.path(DIR1, DIR2, FILE_NAME): for composing paths, no more /\ confusions;
 - normalizePath(RELATIVE_PATH): converts relative path to absolute path;
 - list.files(DIR): lists files in a folder;
 - file.exists(PATH): checks if a file exists at the specified path.

A decorative graphic on the left side of the slide consisting of several overlapping, wavy, grey lines that create a sense of movement and depth.

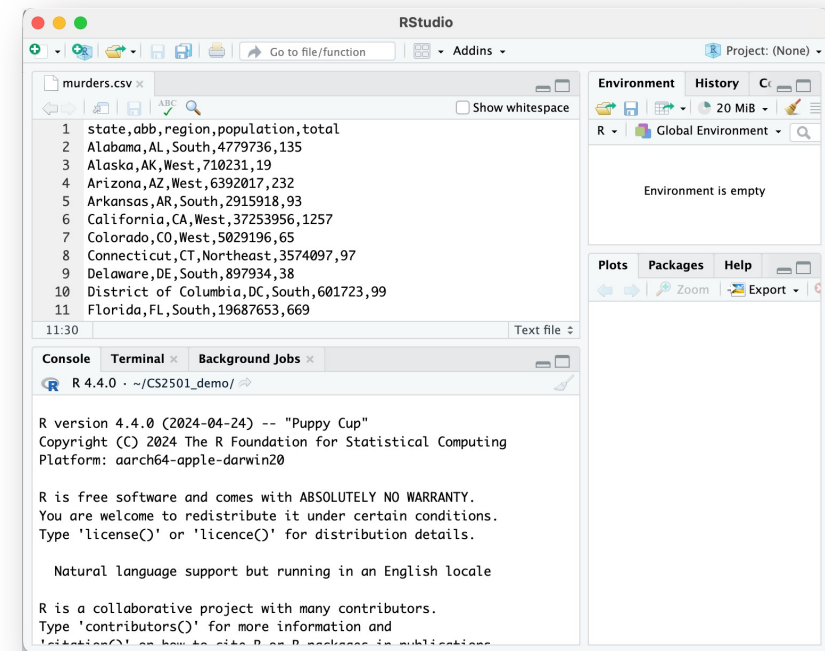
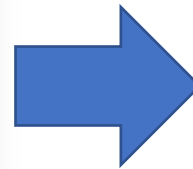
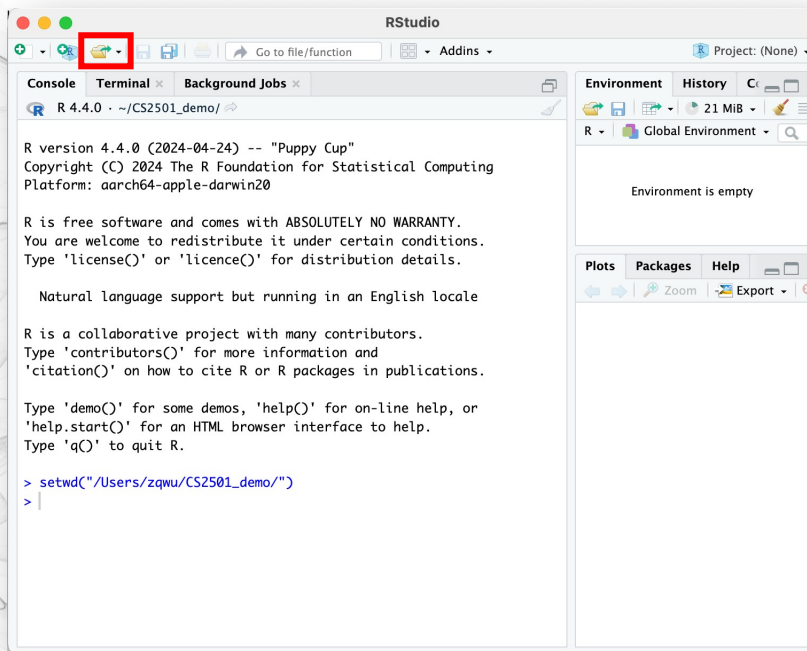
Working Directory

- Relative path is generally more recommended, as absolute paths may vary across devices, platforms, etc., making them less portable
- Preferred approach: organize all your data files, codes, and markdown/notebook files in the same place (i.e., working directory)

```
> getwd()
[1] "/Users/zqwu"
> dir.create("CS2501_demo")
> setwd("/Users/zqwu/CS2501_demo/")
> filename <- "murders.csv"
> dir <- system.file("extdata", package = "dslabs")
> data_file_path <- file.path(dir, filename)
> file.copy(data_file_path, "murders.csv")
[1] TRUE
> list.files()
[1] "murders.csv"
```

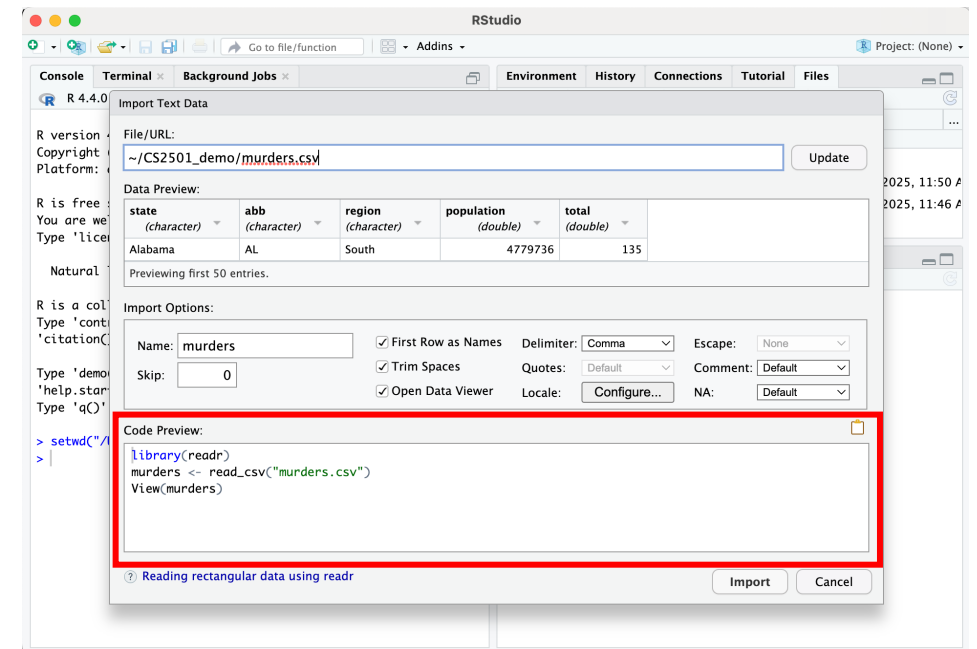
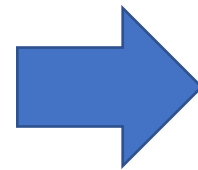
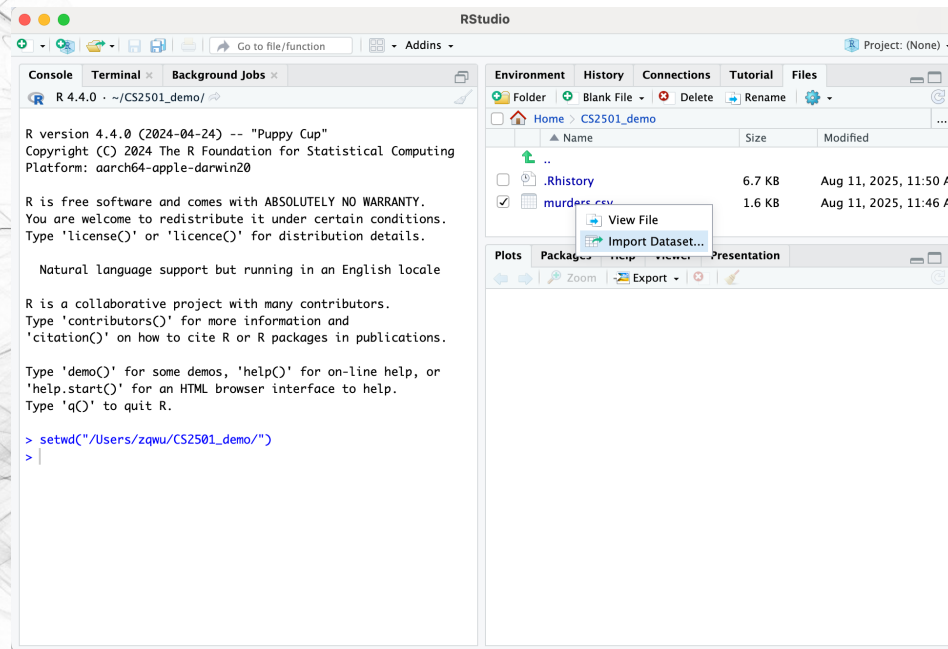
- Use `getwd()`, `setwd(PATH)` to view/change your working directory
- Move your data files to your working directory
- Share the entire working directory when communicating results, data, codes, etc.

Read data file using Rstudio GUI



- This is opening the data file in your editor, not importing it to your environment

Read data file using Rstudio GUI



- Not recommended, use codes when you can



read_csv

```
> library(readr)
> murders <- read_csv("murders.csv")
Rows: 51 Columns: 5
— Column specification —
Delimiter: ","
chr (3): state, abb, region
dbl (2): population, total

i Use `spec()` to retrieve the full column
s data.
i Specify the column types or set `show_col
iet this message.
> head(murders, 3)
# A tibble: 3 × 5
  state   abb region population total
  <chr>  <chr> <chr>      <dbl> <dbl>
1 Alabama AL   South    4779736  135
2 Alaska  AK    West      710231  19
3 Arizona AZ    West    6392017  232
```

```
> spec(murders)
cols(
  state = col_character(),
  abb = col_character(),
  region = col_character(),
  population = col_double(),
  total = col_double()
)
```

- Use `read_csv` to read a local file
- See also `?read_csv` for its optional arguments

read_csv

```
> covid <- read_csv("http://www.bio8.cs.hku.hk/comp2501/covid.csv")
Rows: 47480 Columns: 12
— Column specification —
Delimiter: ","
chr (5): dateRep, countriesAndTerritories, geoId, countryterritoryCode, contine...
dbl (7): day, month, year, cases, deaths, popData2019, Cumulative_number_for_14...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> head(covid)
# A tibble: 6 × 12
  dateRep      day month  year cases deaths countriesAndTerritories geoId
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <chr>                <chr>
1 07/10/2020     7    10  2020     62     2 Afghanistan             AF
2 06/10/2020     6    10  2020    145     5 Afghanistan             AF
3 05/10/2020     5    10  2020     44     0 Afghanistan             AF
4 04/10/2020     4    10  2020      7     4 Afghanistan             AF
5 03/10/2020     3    10  2020      5     0 Afghanistan             AF
6 02/10/2020     2    10  2020     17     0 Afghanistan             AF
# i 4 more variables: countryterritoryCode <chr>, popData2019 <dbl>,
#   continentExp <chr>,
#   `Cumulative_number_for_14_days_of_COVID-19_cases_per_100000` <dbl>
> spec(covid)
cols(
  dateRep = col_character(),
  day = col_double(),
  month = col_double(),
  year = col_double(),
  cases = col_double(),
  deaths = col_double(),
  countriesAndTerritories = col_character(),
  geoId = col_character(),
  countryterritoryCode = col_character(),
  popData2019 = col_double(),
  continentExp = col_character(),
  `Cumulative_number_for_14_days_of_COVID-19_cases_per_100000` = col_double()
)
```

<http://www.bio8.cs.hku.hk/comp2501/covid.csv>

- You can also use `read_csv` to (download and) read a csv file from url
- `download.file(url, "covid.csv")` for explicitly downloading a copy



Other file readers

`read_csv` is in the `readr` library (part of `tidyverse`)

There are some other functions that suit different file format

Function	Format	Typical suffix
<code>read_table</code>	white space separated values	txt
<code>read_csv</code>	comma separated values	csv
<code>read_csv2</code>	semicolon separated values	csv
<code>read_tsv</code>	tab delimited separated values	tsv
<code>read_delim</code>	general text file format, must define delimiter	txt

Hey, how to read Microsoft Excel?





readxl library

Function	Format	Typical suffix
read_excel	auto detect the format	xls, xlsx
read_xls	original format	xls
read_xlsx	new format	xlsx

The Microsoft Excel formats permit you to have more than one spreadsheet in one file, referred to as sheets.

The functions listed above read the **FIRST** sheet by default.

To read other sheets, use names of the sheets for the `sheet` argument.

Read xls and xlsx files

Description

Read xls and xlsx files

`read_excel()` calls `excel_format()` to determine if path is xls or xlsx, based on the file extension and the file itself, in that order. Use `read_xls()` and `read_xlsx()` directly if you know better and want to prevent such guessing.

Usage

```
read_excel(  
  path,  
  sheet = NULL,  
  range = NULL,  
  col_names = TRUE,  
  col_types = NULL,  
  na = "",  
  trim_ws = TRUE,
```



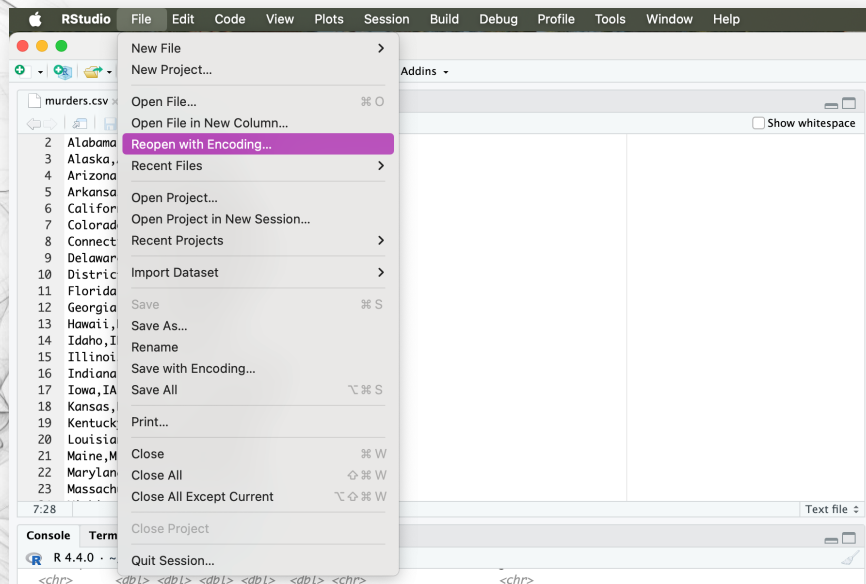
Encoding

- Encoding: text => bit vector

Encoding	Description	Character Set	Size	Use Case
ASCII	Original character encoding standard.	English letters, digits, and basic symbols (128 characters).	1 byte/char	Legacy files, simple text files.
UTF-8	Most widely used Unicode encoding; variable length; backward compatible with ASCII.	All Unicode characters.	1–4 bytes/char	Default for modern systems, web applications, and multilingual text.
UTF-16	Variable-length Unicode encoding; not directly compatible with ASCII.	All Unicode characters.	2 or 4 bytes/char	Multilingual text, applications requiring wide character sets, e.g., Windows APIs.
GBK & GB18030	Simplified and traditional Chinese encoding (GB18030 is a superset of GBK).	~21,000 Chinese characters (GBK); all Unicode characters (GB18030).	2 bytes/char (GBK); 1–4 bytes/char (GB18030).	Simplified Chinese text files (GBK); full Unicode support (GB18030).
Big5 & Big5-HKSCS	Traditional Chinese encoding used in Taiwan, Hong Kong, etc.	~13,000 Traditional Chinese characters (Big5); additional characters in Big5-HKSCS.	2 bytes/char	Traditional Chinese text files.

- RStudio defaults to UTF-8

Encoding



```
> read_csv(file_path)
Rows: 3 Columns: 4
Error in nchar(x, "width") : invalid multibyte string, element 1
> read_csv(file_path, locale = locale(encoding = "GB18030"))
Rows: 3 Columns: 4
— Column specification —
Delimiter: ","
chr (3): 姓, 名, 城市
dbl (1): 年龄

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
# A tibble: 3 × 4
  姓    名    年龄 城市
  <chr> <chr> <dbl> <chr>
1 张    三      25 北京
2 李    四      30 上海
3 王    五      28 广州
```



Good habits of using Excel/Spreadsheet to process your data

- Be Consistent
- Choose Good Names for Things
- Write Dates as YYYY-MM-DD
- No Empty Cells
- Put Just One Thing in a Cell
- Make It a Rectangle
- Create a Data Dictionary
- No Calculations in the Raw Data Files
- Do Not Use Font Color or Highlighting as Data
- Make Backups
- Use Data Validation to Avoid Errors
- Save the Data as Text Files

See also: <https://rafalab.dfci.harvard.edu/dsbook/importing-data.html#organizing-data-with-spreadsheets>